

```
# Creating example documents
doc_1 = "A whopping 96.5 percent of water on Earth is in our oceans, covering 71 percent of the surface of our planet. And at a
doc_2 = "One-third of your life is spent sleeping. Sleeping 7-9 hours each night should help your body heal itself, activate th
doc_3 = "A newborn baby is 78 percent water. Adults are 55-60 percent water. Water is involved in just about everything our bod
doc_4 = "While still in high school, a student went 264.4 hours without sleep, for which he won first place in the 10th Annual
doc_5 = "We experience water in all three states: solid ice, liquid water, and gas water vapor."

# Create corpus
corpus = [doc_1, doc_2, doc_3, doc_4, doc_5]
```

```
import string
import nltk
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('omw-1.4')
from nltk.corpus import stopwords
from nltk.stem.wordnet import WordNetLemmatizer

# remove stopwords, punctuation, and normalize the corpus
stop = set(stopwords.words('english'))
exclude = set(string.punctuation)
lemma = WordNetLemmatizer()

def clean(doc):
    stop_free = " ".join([i for i in doc.lower().split() if i not in stop])
    punc_free = "".join(ch for ch in stop_free if ch not in exclude)
    normalized = " ".join(lemma.lemmatize(word) for word in punc_free.split())
    return normalized

clean_corpus = [clean(doc).split() for doc in corpus]
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data] Package omw-1.4 is already up-to-date!
```

```
!pip install gensim
```

```
Requirement already satisfied: gensim in /usr/local/lib/python3.12/dist-packages (4.3.3)
Requirement already satisfied: numpy<2.0,>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.26.4)
Requirement already satisfied: scipy<1.14.0,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.13.1)
Requirement already satisfied: smart-open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.3.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open>=1.8.1->gensim) (1.17.3)
```

```
from gensim import corpora

# Creating document-term matrix
dictionary = corpora.Dictionary(clean_corpus)
doc_term_matrix = [dictionary.doc2bow(doc) for doc in clean_corpus]
```

Start coding or [generate](#) with AI.

```
from gensim.models import LsiModel

# LSA model
lsa = LsiModel(doc_term_matrix, num_topics=3, id2word = dictionary)

# LSA model
print(lsa.print_topics(num_topics=3, num_words=3))

"""
[
(0, '0.555*"water" + 0.489*"percent" + 0.239*"planet"'),
(1, '0.361*"sleeping" + 0.215*"hour" + 0.215*"still"'),
(2, '-0.562*"water" + 0.231*"rain" + 0.231*"planet"')
]
"""
```

```
[ (0, '0.555*water" + 0.489*percent" + 0.239*planet"'), (1, '0.361*sleeping" + 0.215*still" + 0.215*hour"'), (2, '0.562*water" + 0.489*percent" + 0.239*planet"'), (1, '0.361*sleeping" + 0.215*hour" + 0.215*still"'), (2, '-0.562*water" + 0.231*rain" + 0.231*planet"') ]
```

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```
from gensim.models import LdaModel

# LDA model
lda = LdaModel(doc_term_matrix, num_topics=3, id2word = dictionary)

# Results
print(lda.print_topics(num_topics=3, num_words=3))

"""
[
(0, '0.071*water" + 0.025*state" + 0.025*three"'),
(1, '0.030*still" + 0.028*hour" + 0.026*sleeping"'),
(2, '0.073*percent" + 0.069*water" + 0.031*rain"')
]
"""
```

WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider increasing the number of passes or iterations

```
[ (0, '0.053*percent" + 0.037*planet" + 0.036*water"'), (1, '0.037*water" + 0.028*still" + 0.026*hour"'), (2, '0.088*water" + 0.071*water" + 0.025*state" + 0.025*three"'), (1, '0.030*still" + 0.028*hour" + 0.026*sleeping"'), (2, '0.073*percent" + 0.069*water" + 0.031*rain"') ]
```

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```
!pip install pyLDAvis matplotlib
```

```
Collecting pyLDAvis
  Downloading pyLDAvis-3.4.1-py3-none-any.whl.metadata (4.2 kB)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: numpy>=1.24.2 in /usr/local/lib/python3.12/dist-packages (from pyLDAvis) (1.26.4)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from pyLDAvis) (1.13.1)
Requirement already satisfied: pandas>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from pyLDAvis) (2.2.2)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from pyLDAvis) (1.5.2)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from pyLDAvis) (3.1.6)
Requirement already satisfied: Numexpr in /usr/local/lib/python3.12/dist-packages (from pyLDAvis) (2.11.0)
Collecting funcy (from pyLDAvis)
  Downloading funcy-2.0-py2.py3-none-any.whl.metadata (5.9 kB)
Requirement already satisfied: scikit-learn>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from pyLDAvis) (1.6.1)
Requirement already satisfied: gensim in /usr/local/lib/python3.12/dist-packages (from pyLDAvis) (4.3.3)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from pyLDAvis) (75.2.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.59.2)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.0.0->pyLDAvis) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.0.0->pyLDAvis) (2025.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.0.0->pyLDAvis) (3.5.0)
Requirement already satisfied: smart-open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim->pyLDAvis) (7.3.1)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->pyLDAvis) (3.0.2)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open>=1.8.1->gensim->pyLDAvis) (1.17.0)
Downloading pyLDAvis-3.4.1-py3-none-any.whl (2.6 MB)
2.6/2.6 MB 20.9 MB/s eta 0:00:00
Downloading funcy-2.0-py2.py3-none-any.whl (30 kB)
Installing collected packages: funcy, pyLDAvis
Successfully installed funcy-2.0 pyLDAvis-3.4.1
```

```
import pyLDAvis
import pyLDAvis.gensim_models as gensimvis
import matplotlib.pyplot as plt
from IPython.display import display, HTML

# Prepare LDA visualization
lda_display = gensimvis.prepare(lda, doc_term_matrix, dictionary, sort_topics=False)

# Show interactive visualization (in notebook / Jupyter)
# pyLDAvis.display(lda_display)
display(HTML(pyLDAvis.prepared_data_to_html(lda_display)))

# If you're not in Jupyter (e.g. running .py script),
# you can save as HTML and open in browser:
```

```
pyLDAvis.save_html(lda_display, "lda_visualization.html")
```

Selected Topic:

Previous Topic

Next Topic

Clear Topic

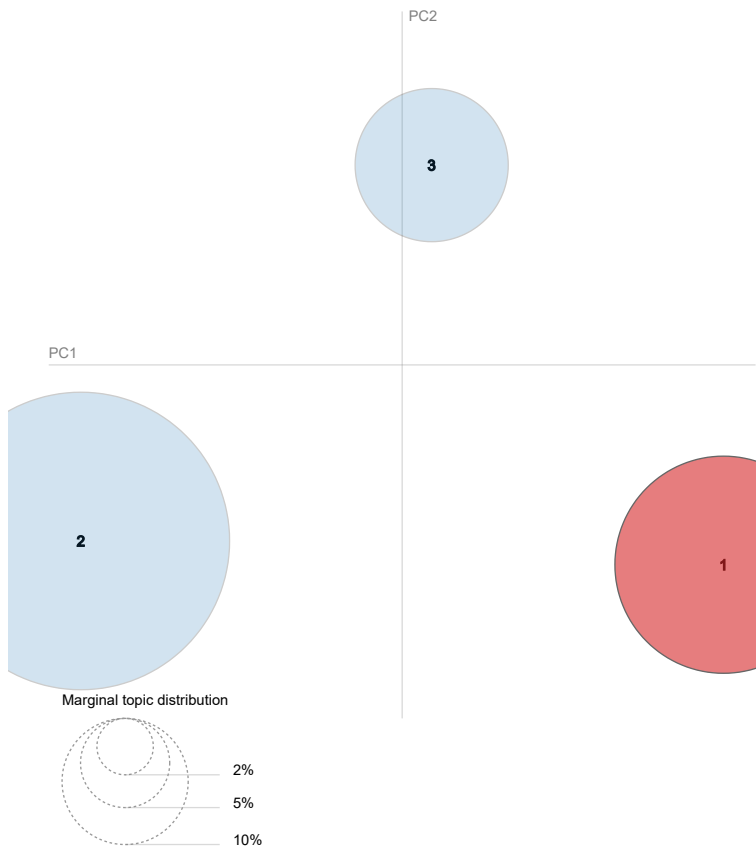
Slide to adjust relevance metric:(2)

$\lambda = 1$

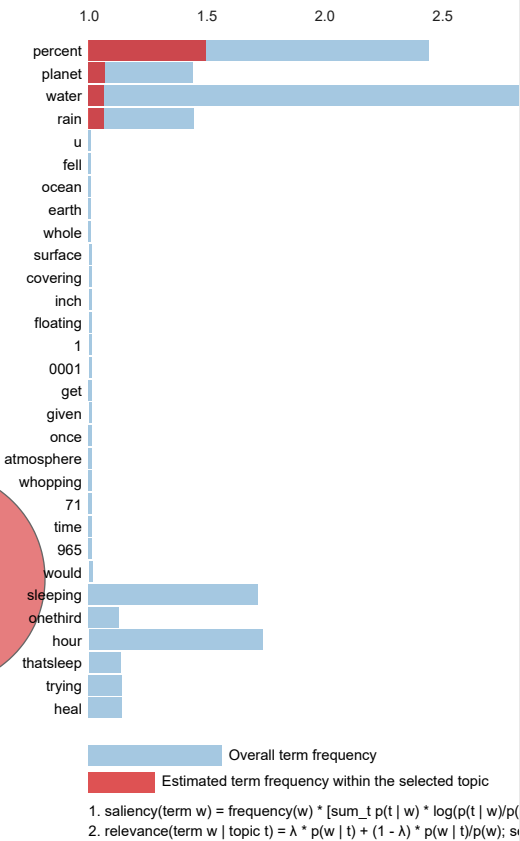
0.0

0.2

Intertopic Distance Map (via multidimensional scaling)



Top-30 Most Relevant Terms for Topic



```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated, use datetime.datetime.now()
return datetime.datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated, use datetime.datetime.now()
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```

